

Doc 1: Exponents, Logarithms, and e

Module 0: Warm-up | Theory recap (read first, about 10 minutes)

1. Exponents: repeated multiplication

a^n means multiply a by itself n times. For example $2^3 = 2 \times 2 \times 2 = 8$. Six rules cover almost everything you will need:

Multiply (same base)	$a^m \times a^n = a^{m+n}$	$2^3 \times 2^2 = 2^5 = 32$
Divide (same base)	$a^m / a^n = a^{m-n}$	$3^5 / 3^3 = 3^2 = 9$
Power of a power	$(a^m)^n = a^{m \times n}$	$(2^3)^2 = 2^6 = 64$
Zero exponent	$a^0 = 1$	$5^0 = 1$
Negative exponent	$a^{-n} = 1 / a^n$	$2^{-3} = 1/8 = 0.125$
Fractional exponent	$a^{1/2} = \text{square root of } a$	$9^{1/2} = 3$

2. Logarithms: the question exponents can't answer

A logarithm asks: **what power do I raise the base to, to get this number?** $\log_b(x) = y$ means $b^y = x$.

Example: $\log_2(8) = 3$ because $2^3 = 8$.

Product to sum	$\log(x \times y) = \log x + \log y$	$\log_2(8 \times 4) = 3 + 2 = 5$
Quotient to difference	$\log(x / y) = \log x - \log y$	$\log_2(8/4) = 3 - 2 = 1$
Power comes down	$\log(x^n) = n \times \log x$	$\log_{10}(100^3) = 3 \times 2 = 6$
Two special values	$\log_b(1) = 0, \log_b(b) = 1$	$\log_5(1) = 0, \log_5(5) = 1$

3. The number e and the natural log

e is a special constant, about **2.718**. The function e^x grows fast and never goes negative. **ln** (natural log) is simply log with base e , so it undoes e^x : $\ln(e^x) = x$ and $e^{\ln x} = x$. Also $\ln(1) = 0$ and $e^0 = 1$.

4. Why machine learning needs all this

- **Sigmoid** squashes any number into (0, 1): $\text{sigmoid}(x) = 1 / (1 + e^{-x})$
- **Softmax** turns scores into probabilities: $\text{softmax}(z_i) = e^{z_i} / (\text{sum of all } e^z)$
- **Cross-entropy loss** punishes wrong confidence: $\text{loss} = -\ln(\text{probability given to the correct class})$
- **Log-likelihood**: $\ln$ turns a long product of probabilities into a sum, which is easier and safer to compute.

Cheat sheet (values used in the exercises)

x	-2	-1	0	1	2	3
e^x	0.135	0.368	1	2.718	7.389	20.086
p	0.1	0.2	0.5	0.7	0.9	
$-\ln(p)$	2.303	1.609	0.693	0.357	0.105	

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Worked examples: every cell filled in. Read them, then cover the answers and redo them.

Example 1: Sigmoid on a grid

Input x	-1	0	1
Step 1: e^{-x}	2.718	1.000	0.368
Step 2: $1 + e^{-x}$	3.718	2.000	1.368
Step 3: $1 / \text{step 2}$	0.269	0.500	0.731

Notice: big negative x gives a value near 0, x = 0 gives exactly 0.5, big positive x gives a value near 1.

Example 2: Softmax on a grid

Scores z	1	2	0
Step 1: e^z	2.718	7.389	1.000
Step 2: sum of row 2	11.107		
Step 3: each e^z / sum	0.245	0.665	0.090
Check: add probabilities	1.000		

The biggest score (2) gets the biggest probability. The probabilities always add up to 1.

Example 3: Cross-entropy loss

If the model gives probability p to the **correct** class, the loss is $-\ln(p)$. Read the values from the cheat sheet:

Probability p on the correct class	0.9	0.5	0.2
Loss = $-\ln(p)$	0.105	0.693	1.609

More confident and right means a small loss. Confident and wrong means a big loss.

Example 4: Logs turn products into sums

Three independent events have probabilities 0.5, 0.5 and 0.2. Product: $0.5 \times 0.5 \times 0.2 = 0.05$. Using $\ln$ values instead: $\ln(0.5) + \ln(0.5) + \ln(0.2) = -0.693 - 0.693 - 1.609 = -2.995$, which matches $\ln(0.05)$, about -3.0. With thousands of probabilities, the raw product becomes far too tiny for a computer, so ML uses the sum of logs.

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Exercises: use a pencil. Yellow cells are yours to fill in. Name: _____ Date: _____

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p	0.1	0.2	0.5	0.7	0.9	
$-\ln(p)$	2.303	1.609	0.693	0.357	0.105	

A. Exponents			B. Logarithms		
1.	$2^3 \times 2^2$		1.	$\log_2(16)$	
2.	$3^5 / 3^3$		2.	$\log_{10}(1000)$	
3.	$(2^3)^2$		3.	$\log_3(1)$	
4.	5^0		4.	$\log_5(5)$	
5.	2^{-3}		5.	$\log_2(1/4)$	
6.	$9^{1/2}$		6.	$\ln(e^3)$	
7.	$10^3 \times 10^{-1}$		7.	$\log_2(8) + \log_2(4)$	
8.	$4^{3/2}$ (hint: square root first)		8.	$\log_{10}(100^3)$	

C. Sigmoid grid (use the cheat sheet for e^x)

Input x	-2	0	1	2
e^{-x}				
$1 + e^{-x}$				
$1 / (1 + e^{-x})$				

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Exercises, continued

D. Softmax grid

Scores z	2	1	0	-1
Step 1: e^z				
Step 2: sum				
Step 3: probabilities				
Check: total				

Quick think: if the scores are 5, 5, 5, what probability does each get? _____ (no calculation needed)

E. Cross-entropy loss

Probability p on the correct class	0.7	0.2	0.9	0.1
Loss = $-\ln(p)$				

Which case has the biggest loss? _____ Why does that make sense?

F. Challenge: products to sums

Four independent events have probabilities 0.9, 0.5, 0.2, 0.1. Fill in the grid, then compare the sum of the logs with what you'd expect.

Probability p	0.9	0.5	0.2	0.1
$\ln(p) = -(\text{value from cheat sheet})$				
Sum of $\ln$ values				

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Answer key: check your work only after finishing

A. Exponents			B. Logarithms		
1.	$2^3 \times 2^2$	32	1.	$\log_2(16)$	4
2.	$3^5 / 3^3$	9	2.	$\log_{10}(1000)$	3
3.	$(2^3)^2$	64	3.	$\log_3(1)$	0
4.	5^0	1	4.	$\log_5(5)$	1
5.	2^{-3}	0.125	5.	$\log_2(1/4)$	-2
6.	$9^{1/2}$	3	6.	$\ln(e^3)$	3
7.	$10^3 \times 10^{-1}$	100	7.	$\log_2(8) + \log_2(4)$	5
8.	$4^{3/2}$ (hint: square root first)	8	8.	$\log_{10}(100^3)$	6

C. Sigmoid

Input x	-2	0	1	2
e^{-x}	7.389	1.000	0.368	0.135
$1 + e^{-x}$	8.389	2.000	1.368	1.135
$1 / (1 + e^{-x})$	0.119	0.500	0.731	0.881

D. Softmax

Scores z	2	1	0	-1
e^z	7.389	2.718	1.000	0.368
Sum	11.475			
Probabilities	0.644	0.237	0.087	0.032
Check: total	1.000			

Quick think: equal scores give equal probabilities, so 1/3 each (about 0.333).

E and F

p on correct class	0.7	0.2	0.9	0.1
Loss = $-\ln(p)$	0.357	1.609	0.105	2.303

Biggest loss: $p = 0.1$. The model was confident in the wrong answers and gave the right one almost no probability.

Probability p	0.9	0.5	0.2	0.1
$\ln(p)$	-0.105	-0.693	-1.609	-2.303
Sum of $\ln$ values	-4.710			

Product $0.9 \times 0.5 \times 0.2 \times 0.1 = 0.009$, and $\ln(0.009)$ is about -4.71, matching the sum of the logs.

Key takeaways: exponent rules add and multiply powers; log undoes exponent; e^x is always positive; sigmoid and softmax are built from e^x ; cross-entropy is $-\ln(p)$; logs turn products into sums.

Next: Doc 2 - Functions, graphs, slopes and Sigma notation.